



EDA_project.py X

C: > Users > Admin > Downloads > EDA_project.py > ...

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1  import pandas as pd
2  import matplotlib.pyplot as plt
3  import seaborn as sns
4  df = pd.read_excel(r"C:\Users\Admin\Downloads\Dataset for Data Analytics.xlsx")
5  print("First 5 Rows:")
6  print(df.head())
7  print("\nDataset Shape:")
8  print(df.shape)
9  print("\nMissing Values Before Cleaning:")
10 print(df.isnull().sum())
11 df["CouponCode"] = df["CouponCode"].fillna("No Coupon")
12 print("\nMissing Values After Cleaning:")
13 print(df.isnull().sum())
14 print("\nBasic Statistics:")
15 print(df.describe())
16 print("\nMean Values:")
17 print(df[["Quantity", "UnitPrice", "ItemsInCart", "TotalPrice"]].mean())
18 print("\nMedian Values:")
19 print(df[["Quantity", "UnitPrice", "ItemsInCart", "TotalPrice"]].median())
20 product_sales = df.groupby("Product")["TotalPrice"].sum()
21 plt.figure(figsize=(10, 5))
22 product_sales.plot(kind="bar")
23 plt.title("Total Sales by Product")
24 plt.xlabel("Product")
25 plt.ylabel("Total Sales")
26 plt.xticks(rotation=45)
27 plt.tight_layout()
28 plt.show()
29 plt.figure(figsize=(8, 5))
30 sns.histplot(df["TotalPrice"], bins=25, kde=True)
31 plt.title("Distribution of Total Price")
32 plt.xlabel("Total Price")
33 plt.ylabel("Frequency")
34 plt.tight_layout()
35 plt.show()
36 df.to_excel("cleaned_dataset.xlsx", index=False)
37 print("\nEDA Project Completed Successfully!")

```

